

Grade 8
Unit 3
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

1. Complete the statement.

When rewriting 7,000 in scientific notation, the factor 7 is

1 and

- ☒ less than
☐ greater than or equal to

10.

- ☐ less than
☒ greater than or equal to

The digit 7 is in the thousands place. A thousand expressed as a power of 10 is

- ☒ 10^3 .
☐ 10^4 .

Therefore, 7,000 is written as

- ☒ 7×10^3
☐ 7×10^4

in scientific notation.

2. Complete the table.

Scientific Notation	Standard Form
4.6×10^6	4,600,000
2.3×10^6	2,300,000
6.9×10^6	6,900,000

3. How many times larger is 4.6×10^6 than 2.3×10^6 ?

$$\frac{4.6}{2.3} \times \frac{10^6}{10^6} = 2 \times 1 = 2 \text{ times larger}$$

4. How many times smaller is 2.3×10^6 than 6.9×10^6 ?

$$\frac{6.9}{2.3} \times \frac{10^6}{10^6} = 3 \times 1 = 3 \text{ times smaller}$$

5. Consider the expression 88×10^7 . The expression is also equivalent to 88,000,000. Use the definition of scientific notation to explain why 88×10^7 is not the correct expression of 88,000,000 in scientific notation.

88 is not greater than or equal to 1 and less than 10.

6. Complete the table.

Scientific Notation	Standard Form
4.1×10^5	410,000
8.2×10^6	8,200,000
1.64×10^7	16,400,000

7. How many times larger is 8.2 than 4.1?

2

8. How many times larger is 10^7 than 10^5 ?

100

9. How many times larger is 8.2×10^7 than 4.1×10^5 ?

2

×

100

= 200 times larger

10. The population of visitors going to Disney World was about 6.94×10^6 in 2020. The population of visitors going to Universal Studios in 2020 was about 1.7×10^6 .

Complete the statement.

In 2020, the population of visitors to Disney World was about 4 times larger than the population of visitors going to Universal Studios.

$$\frac{6.94}{1.7} \times \frac{10^6}{10^6}$$

$$\approx 4 \times 1$$